

Practical 2

Aim:

Hadoop Installation, Configuration and HDFS Operations on Windows/Linux

Problem definition:

You have recently joined DataAnalytics Inc. as a Hadoop Administrator. The organization processes large-scale customer, sales, and marketing data using the Hadoop ecosystem. Due to increasing data volumes, the company has decided to migrate its data storage infrastructure from traditional file systems to Hadoop Distributed File System (HDFS).

As a Hadoop Administrator, your responsibility is to install and configure Hadoop in a Single-Node Cluster environment on Windows/Linux, verify the Hadoop services, manage HDFS storage, and perform data management operations.

Management wants a fully functional Hadoop environment that can store, retrieve, and process big data efficiently.

Your task is to install Hadoop, configure the Hadoop ecosystem, start Hadoop services, and perform HDFS operations using Hadoop commands.

Objectives:

- Install and configure Hadoop in a single-node environment.
- Verify Hadoop installation and services.
- Perform HDFS file and directory operations.
- Monitor Hadoop cluster and storage utilization.
- Understand the working of HDFS.

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Output :

- Hadoop Installation & Configuration

```
Terminal Shell Edit View Window Help
patelkrina -- zsh -- 207x49

Last login: Fri Aug 7 13:17:27 on console
(base) patelkrina@patels-MacBook-Air-4 ~ % java -version
echo
java version "24.0.2" 2025-07-15
Java(TM) SE Runtime Environment (build 24.0.2+12-54)
Java HotSpot(TM) 64-Bit Server VM (build 24.0.2+12-54, mixed mode, sharing)

(base) patelkrina@patels-MacBook-Air-4 ~ % hadoop version
echo
Hadoop 3.5.0
Source code repository https://github.com/apache/hadoop.git -r dbcc7cd77180eeb32cd84f85b53a5193a5f9af0
Compiled by cnauroth on 2026-03-24T16:56Z
Compiled on platform linux-x86_64
Compiled with protoc 3.25.5
From source with checksum 64582ed8a62a169c39a19c734a0f33b5
This command was run using /opt/homebrew/Cellar/hadoop/3.5.0/libexec/share/hadoop/common/hadoop-common-3.5.0.jar
```

- Verify Running Hadoop Processes.

```
Terminal Shell Edit View Window Help
patelkrina -- zsh -- 207x8

(base) patelkrina@patels-MacBook-Air-4 ~ % jps
echo
1364 NameNode
2845 Jps
1469 DataNode
1615 SecondaryNameNode

(base) patelkrina@patels-MacBook-Air-4 ~ %
```

- Create DATASET

```
Terminal Shell Edit View Window Help
patelkrina -- zsh -- 207x19

(base) patelkrina@patels-MacBook-Air-4 ~ % cd ~
echo

(base) patelkrina@patels-MacBook-Air-4 ~ % cat > sales_data.txt
echo
ProductA 120
ProductB 80
ProductC 150
ProductA 200
ProductB 900
ProductB 900
(base) patelkrina@patels-MacBook-Air-4 ~ % cat sales_data.txt
echo
ProductA 120
ProductB 80
ProductC 150
ProductA 200
ProductB 90
(base) patelkrina@patels-MacBook-Air-4 ~ %
```

1. Create directories in HDFS.

```
Terminal Shell Edit View Window Help
patelkrina --zsh - 207x17

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -mkdir /MarketingAnalytics 2>/dev/null
echo

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -mkdir /MarketingAnalytics/RawData 2>/dev/null
echo

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -mkdir /MarketingAnalytics/Reports 2>/dev/null
echo

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -ls /MarketingAnalytics 2>/dev/null
echo
Found 2 items
drwxr-xr-x - patelkrina supergroup      0 2026-08-07 10:17 /MarketingAnalytics/RawData
drwxr-xr-x - patelkrina supergroup      0 2026-08-07 10:12 /MarketingAnalytics/Reports
(base) patelkrina@patels-MacBook-Air-4 ~ %
```

2. Upload files to HDFS.

```
Terminal Shell Edit View Window Help
patelkrina --zsh - 207x9

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -put sales_data.txt /MarketingAnalytics/RawData/ 2>/dev/null
echo

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -ls /MarketingAnalytics/RawData 2>/dev/null
echo
Found 1 items
-rw-r--r--  1 patelkrina supergroup      63 2026-08-07 10:17 /MarketingAnalytics/RawData/sales_data.txt
(base) patelkrina@patels-MacBook-Air-4 ~ %
```

3. List HDFS files and directories.

```
Terminal Shell Edit View Window Help
patelkrina --zsh - 207x18

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -ls / 2>/dev/null
echo
Found 2 items
drwxr-xr-x - patelkrina supergroup      0 2026-08-07 10:12 /MarketingAnalytics
drwxr-xr-x - patelkrina supergroup      0 2026-08-06 13:28 /test

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -ls /MarketingAnalytics 2>/dev/null
echo
Found 2 items
drwxr-xr-x - patelkrina supergroup      0 2026-08-07 10:17 /MarketingAnalytics/RawData
drwxr-xr-x - patelkrina supergroup      0 2026-08-07 10:12 /MarketingAnalytics/Reports

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -ls /MarketingAnalytics/RawData 2>/dev/null
echo
Found 1 items
-rw-r--r--  1 patelkrina supergroup      63 2026-08-07 10:17 /MarketingAnalytics/RawData/sales_data.txt
(base) patelkrina@patels-MacBook-Air-4 ~ %
```

4. Display file contents.

```
Terminal Shell Edit View Window Help
patelkrina --zsh - 207x10

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -cat /MarketingAnalytics/RawData/sales_data.txt 2>/dev/null
echo
ProductA 120
ProductB 80
ProductC 150
ProductA 200
ProductB 90
(base) patelkrina@patels-MacBook-Air-4 ~ %
```

5. Copy files within HDFS.

```
Terminal Shell Edit View Window Help
patelkrina --zsh - 207x10

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -cp /MarketingAnalytics/RawData/sales_data.txt /MarketingAnalytics/Reports/ 2>/dev/null
echo

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -ls /MarketingAnalytics/Reports 2>/dev/null
echo
Found 1 items
-rw-r--r--  1 patelkrina supergroup        63 2026-08-07 13:27 /MarketingAnalytics/Reports/sales_data.txt

(base) patelkrina@patels-MacBook-Air-4 ~ %
```

6. Move files within HDFS.

```
Terminal Shell Edit View Window Help
patelkrina --zsh - 207x12

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -mv /MarketingAnalytics/Reports/sales_data.txt /MarketingAnalytics/ 2>/dev/null
echo

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -ls /MarketingAnalytics 2>/dev/null
echo
Found 3 items
drwxr-xr-x  - patelkrina supergroup      0 2026-08-07 10:17 /MarketingAnalytics/RawData
drwxr-xr-x  - patelkrina supergroup      0 2026-08-07 13:27 /MarketingAnalytics/Reports
-rw-r--r--  1 patelkrina supergroup      63 2026-08-07 13:27 /MarketingAnalytics/sales_data.txt

(base) patelkrina@patels-MacBook-Air-4 ~ %
```

7. Rename files.

```
Terminal Shell Edit View Window Help
patelkrina --zsh - 207x12

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -mv /MarketingAnalytics/sales_data.txt /MarketingAnalytics/final_sales.txt 2>/dev/null
echo

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -ls /MarketingAnalytics 2>/dev/null
echo
Found 3 items
drwxr-xr-x  - patelkrina supergroup      0 2026-08-07 10:17 /MarketingAnalytics/RawData
drwxr-xr-x  - patelkrina supergroup      0 2026-08-07 13:27 /MarketingAnalytics/Reports
-rw-r--r--  1 patelkrina supergroup      63 2026-08-07 13:27 /MarketingAnalytics/final_sales.txt

(base) patelkrina@patels-MacBook-Air-4 ~ %
```

8. Download files from HDFS.

```
Terminal Shell Edit View Window Help
patelkrina --zsh - 207x12

(base) patelkrina@patels-MacBook-Air-4 ~ % mkdir -p DownloadsHDFS
echo

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -get /MarketingAnalytics/final_sales.txt DownloadsHDFS/ 2>/dev/null
echo

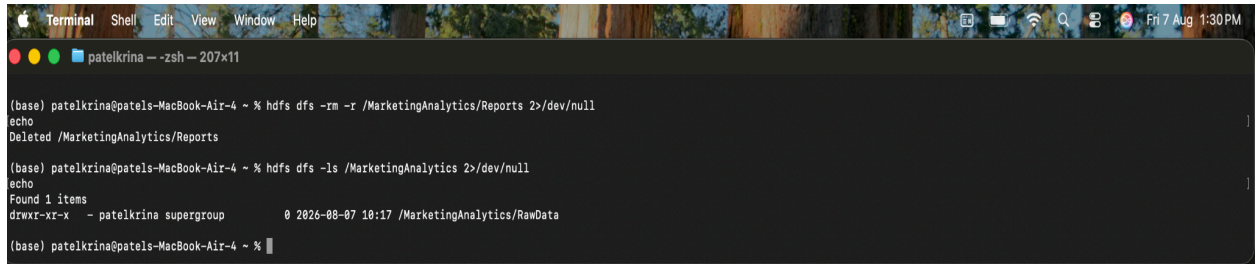
(base) patelkrina@patels-MacBook-Air-4 ~ % ls DownloadsHDFS
echo
final_sales.txt

(base) patelkrina@patels-MacBook-Air-4 ~ %
```

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9. Delete files and directories.

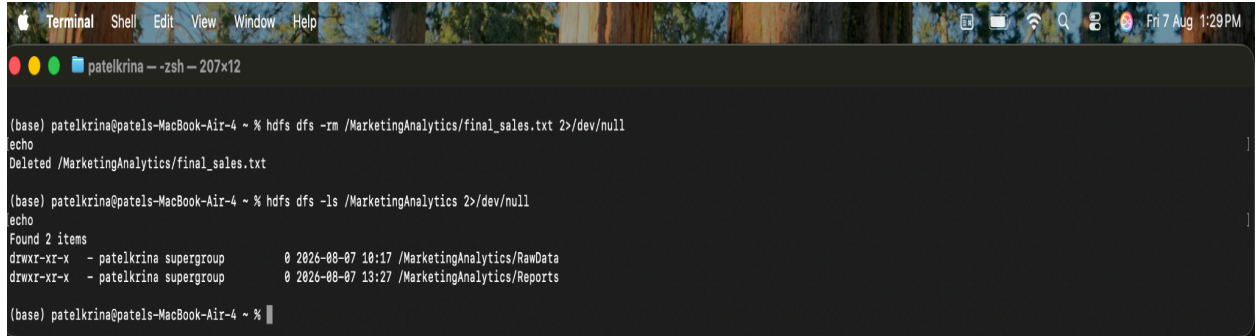


```
Terminal Shell Edit View Window Help
patelkrina ~ -zsh - 207x11

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -rm -r /MarketingAnalytics/Reports 2>/dev/null
echo
Deleted /MarketingAnalytics/Reports

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -ls /MarketingAnalytics 2>/dev/null
echo
Found 1 items
drwxr-xr-x  - patelkrina supergroup          0 2026-08-07 18:17 /MarketingAnalytics/RawData

(base) patelkrina@patels-MacBook-Air-4 ~ %
```



```
Terminal Shell Edit View Window Help
patelkrina ~ -zsh - 207x12

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -rm /MarketingAnalytics/final_sales.txt 2>/dev/null
echo
Deleted /MarketingAnalytics/final_sales.txt

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -ls /MarketingAnalytics 2>/dev/null
echo
Found 2 items
drwxr-xr-x  - patelkrina supergroup          0 2026-08-07 18:17 /MarketingAnalytics/RawData
drwxr-xr-x  - patelkrina supergroup          0 2026-08-07 13:27 /MarketingAnalytics/Reports

(base) patelkrina@patels-MacBook-Air-4 ~ %
```

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10. Check HDFS report.

```
Terminal Shell Edit View Window Help
patelkrina --zsh - 207x49

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfsadmin -report 2>/dev/null
echo
Configured Capacity: 245187195904 (228.27 GB)
Present Capacity: 23338106880 (21.74 GB)
DFS Remaining: 23338086400 (21.74 GB)
DFS Used: 20480 (20 KB)
DFS Used%: 0.00%
Replicated Blocks:
    Under replicated blocks: 0
    Blocks with corrupt replicas: 0
    Missing blocks: 0
    Missing blocks (with replication factor 1): 0
    Badly Distributed Blocks: 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0
Erasure Coded Block Groups:
    Low redundancy block groups: 0
    Block groups with corrupt internal blocks: 0
    Missing block groups: 0
    Badly Distributed Blocks: 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0

-----
Live datanodes (1):
Name: 127.0.0.1:9866 (localhost)
Hostname: localhost
Decommission Status : Normal
Configured Capacity: 245187195904 (228.27 GB)
DFS Used: 20480 (20 KB)
Non DFS Used: 221769089024 (206.54 GB)
DFS Remaining: 23338086400 (21.74 GB)
DFS Used%: 0.00%
DFS Remaining%: 9.52%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Fri Aug 07 13:30:14 IST 2026
Last Block Report: Fri Aug 07 13:20:22 IST 2026
Num of Blocks: 1

(base) patelkrina@patels-MacBook-Air-4 ~ %
```

11. Verify block information.

```
Terminal Shell Edit View Window Help
patelkrina --zsh - 207x57

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs fsck / 2>/dev/null
echo
FSCK started by patelkrina (auth:SIMPLE) from /127.0.0.1 for path / at Fri Aug 07 13:30:43 IST 2026

Status: HEALTHY
Number of data-nodes: 1
Number of racks: 1
Total dirs: 4
Total symlinks: 0

Replicated Blocks:
Total size: 63 B
Total files: 1
Total blocks (validated): 1 (avg. block size 63 B)
Minimally replicated blocks: 1 (100.0 %)
Over-replicated blocks: 0 (0.0 %)
Under-replicated blocks: 0 (0.0 %)
Mis-replicated blocks: 0 (0.0 %)
Default replication factor: 1
Average block replication: 1.0
Missing blocks: 0
Corrupt blocks: 0
Missing replicas: 0 (0.0 %)
Blocks queued for replication: 0

Erasure Coded Block Groups:
Total size: 0 B
Total files: 0
Total block groups (validated): 0
Minimally erasure-coded block groups: 0
Over-erasure-coded block groups: 0
Under-erasure-coded block groups: 0
Unsatisfactory placement block groups: 0
Average block group size: 0.0
Missing block groups: 0
Corrupt block groups: 0
Missing internal blocks: 0
Blocks queued for replication: 0
FSCK ended at Fri Aug 07 13:30:43 IST 2026 in 3 milliseconds

The filesystem under path '/' is HEALTHY
```

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12. Check Hadoop cluster health.

```
Terminal Shell Edit View Window Help
patelkrina -- zsh -- 207x49

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfsadmin -report 2>/dev/null
echo
Configured Capacity: 245187195904 (228.27 GB)
Present Capacity: 23338864480 (21.74 GB)
DFS Remaining: 23338864480 (21.74 GB)
DFS Used: 20480 (20 KB)
DFS Used%: 0.00%
Replicated Blocks:
  Under replicated blocks: 0
  Blocks with corrupt replicas: 0
  Missing blocks: 0
  Missing blocks (with replication factor 1): 0
  Badly Distributed Blocks: 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0
Erasure Coded Block Groups:
  Low redundancy block groups: 0
  Block groups with corrupt internal blocks: 0
  Missing block groups: 0
  Badly Distributed Blocks: 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0

-----
Live datanodes (1):
Name: 127.0.0.1:9866 (localhost)
Hostname: localhost
Decommission Status : Normal
Configured Capacity: 245187195904 (228.27 GB)
DFS Used: 20480 (20 KB)
Non DFS Used: 221709089920 (206.54 GB)
DFS Remaining: 23338864480 (21.74 GB)
DFS Used%: 0.00%
DFS Remaining%: 9.52%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceiver: 0
Last contact: Fri Aug 07 13:30:14 IST 2026
Last Block Report: Fri Aug 07 13:28:22 IST 2026
Num of Blocks: 1

(base) patelkrina@patels-MacBook-Air-4 ~ %
```

13. Monitor Hadoop processes.

```
Terminal Shell Edit View Window Help
patelkrina -- zsh -- 207x8

(base) patelkrina@patels-MacBook-Air-4 ~ % jps
echo
1364 NameNode
2845 Jps
1449 DataNode
1615 SecondaryNameNode

(base) patelkrina@patels-MacBook-Air-4 ~ %
```

14. Check storage utilization.

```
Terminal Shell Edit View Window Help
patelkrina -- zsh -- 207x11

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -du -h / 2>/dev/null
echo
63 63 /MarketingAnalytics
0 0 /test

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -df -h 2>/dev/null
echo
Filesystem          Size Used Available Use%
hdfs://localhost:9000 228.3 G 20 K 21.7 G 0%
```

Result :

Hadoop was successfully configured in a single-node environment. HDFS operations including directory creation, file upload, file listing, file display, copy, move, rename, download, deletion, storage utilization, and cluster monitoring were performed successfully. The Hadoop services were verified and HDFS was functioning correctly.

Key Questions /Analysis:

Q1. Why is Hadoop preferred for Big Data storage?

Answer: Hadoop can store and process very large amounts of data across multiple computers. It is scalable, fault-tolerant, and cost-effective.

Q2. What is the role of NameNode and DataNode in HDFS?

Answer:

- NameNode: Stores metadata and manages the file system.
- DataNode: Stores the actual data blocks and communicates with the NameNode.

Q3. What is the difference between HDFS and traditional file systems?

HDFS	Traditional File System
Distributed storage	Single machine storage
Handles large datasets	Best for small files
Fault tolerant	Limited fault tolerance
Highly scalable	Limited scalability

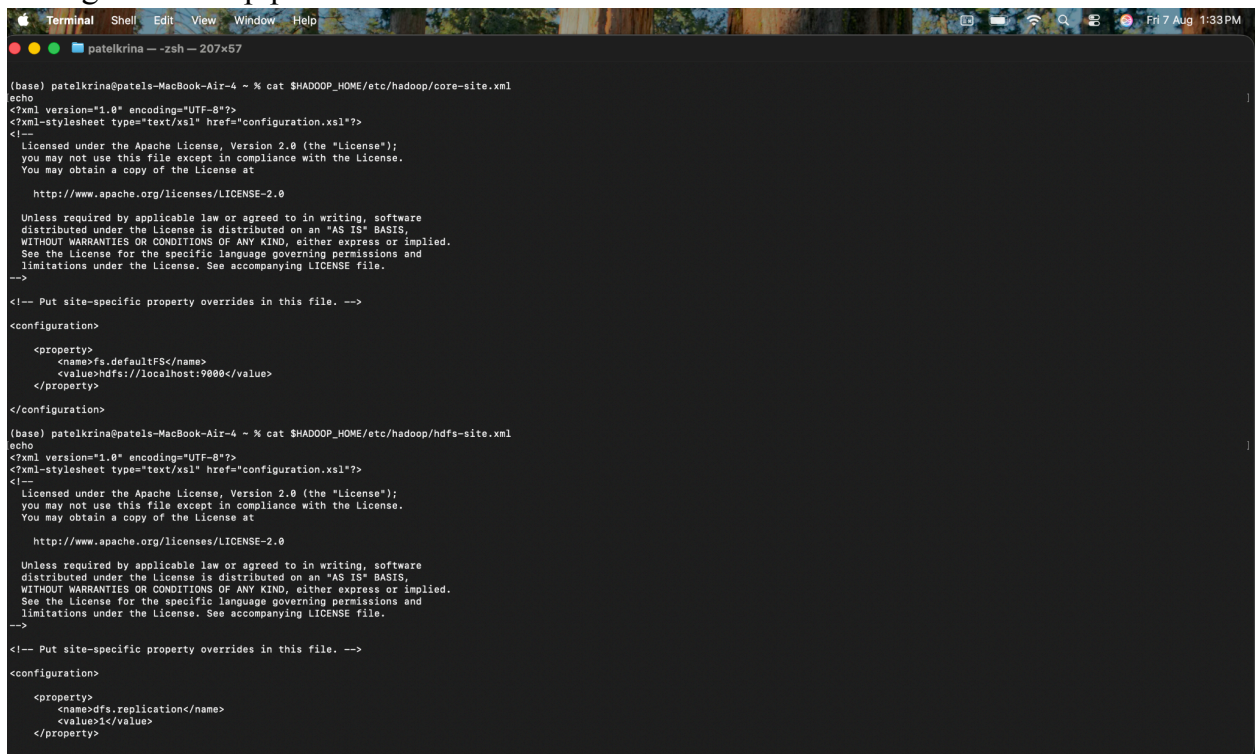
Q4. What are the advantages of distributed storage?

Answer:

- High availability
- Fault tolerance
- Easy scalability
- Faster data processing
- Reliable storage

Supplementary Problems:

1. Configure Hadoop pseudo-distributed mode.



```
(base) patelkrina@patels-MacBook-Air-4 ~ % cat $HADOOP_HOME/etc/hadoop/core-site.xml
<?xml version="1.0" encoding="UTF-8"?>
<!--
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you may not use this file except in compliance with the License.
You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the license for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
-->
<!-- Put site-specific property overrides in this file. -->
<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://localhost:9000</value>
  </property>
</configuration>

(base) patelkrina@patels-MacBook-Air-4 ~ % cat $HADOOP_HOME/etc/hadoop/hdfs-site.xml
<?xml version="1.0" encoding="UTF-8"?>
<!--
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you may not use this file except in compliance with the License.
You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

Unless required by applicable law or agreed to in writing, software
distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the license for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
-->
<!-- Put site-specific property overrides in this file. -->
<configuration>
  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>
</configuration>
```

2. Change HDFS replication factor and observe results.

```
Terminal Shell Edit View Window Help
patelkrina ~ - ssh - 205x53

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -setrep -w 2 /MarketingAnalytics/RawData/sales_data.txt 2>/dev/null
echo
Replication 2 set: /MarketingAnalytics/RawData/sales_data.txt
Waiting for /MarketingAnalytics/RawData/sales_data.txt .....
(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs fsck /MarketingAnalytics/RawData/sales_data.txt -files -blocks 2>/dev/null
echo
FSCK started by patelkrina (auth:SIMPLE) from /192.0.0.1 for path /MarketingAnalytics/RawData/sales_data.txt at Fri Aug 07 14:24:55 IST 2026
/MarketingAnalytics/RawData/sales_data.txt 63 bytes, replicated: replication=2, 1 block(s): Under replicated BP-453149560-127.0.0.1-1786002164770:blk_1073741825_1001. Target Replicas is 2 but found 1 live
replica(s), 0 decommissioned replica(s), 0 decommissioning replica(s).
0 BP-453149560-127.0.0.1-1786002164770:blk_1073741825_1001 len=63 live_rep=1

Status: HEALTHY
Number of data-nodes: 1
Number of racks: 1
Total dirs: 0
Total symlinks: 0
Replicated blocks:
Total size: 63 B
Total blocks (validated): 1 (avg. block size 63 B)
Minimally replicated blocks: 1 (100.0 %)
Over-replicated blocks: 0 (0.0 %)
Under-replicated blocks: 1 (100.0 %)
Mis-replicated blocks: 0 (0.0 %)
Default replication factor: 1
Average block replication: 1.0
Missing blocks: 0
Corrupt blocks: 0
Missing replicas: 1 (50.0 %)
Blocks queued for replication: 0
Erasure Coded Block Groups:
Total size: 0 B
Total block groups (validated): 0
Minimally erasure-coded block groups: 0
Over-erasure-coded block groups: 0
Under-erasure-coded block groups: 0
Unsatisfactory placement block groups: 0
Average block group size: 0 B
Missing block groups: 0
Corrupt block groups: 0
Missing internal blocks: 0
Blocks queued for replication: 0
FSCK ended at Fri Aug 07 14:24:55 IST 2026 in 3 milliseconds

The filesystem under path '/MarketingAnalytics/RawData/sales_data.txt' is HEALTHY
(base) patelkrina@patels-MacBook-Air-4 ~ %
```

3. Upload a large dataset into HDFS.

```
Terminal Shell Edit View Window Help
patelkrina ~ - ssh - 205x21

The filesystem under path '/MarketingAnalytics/RawData/sales_data.txt' is HEALTHY
(base) patelkrina@patels-MacBook-Air-4 ~ % yes "ProductA 120" | head -10000 > large_sales_data.txt
echo
(base) patelkrina@patels-MacBook-Air-4 ~ % ls -lh large_sales_data.txt
echo
-rw-r--r-- 1 patelkrina staff 127K 7 Aug 14:26 large_sales_data.txt
(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -put large_sales_data.txt /MarketingAnalytics/RawData/ 2>/dev/null
echo
(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -ls -h /MarketingAnalytics/RawData 2>/dev/null
echo
Found 2 items
-rw-r--r-- 1 patelkrina supergroup 127.0 K 2026-08-07 14:26 /MarketingAnalytics/RawData/large_sales_data.txt
-rw-r--r-- 2 patelkrina supergroup 63 2026-08-07 10:17 /MarketingAnalytics/RawData/sales_data.txt
(base) patelkrina@patels-MacBook-Air-4 ~ %
```

4. Analyze HDFS storage utilization.

```

patelkrina ~ - ssh - 207x64

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -ls -h /MarketingAnalytics/RawData 2>/dev/null
echo
Found 2 items
-rw-r--r-- 1 patelkrina supergroup 127.0 K 2026-08-07 14:26 /MarketingAnalytics/RawData/large_sales_data.txt
-rw-r--r-- 2 patelkrina supergroup 63 2026-08-07 10:17 /MarketingAnalytics/RawData/sales_data.txt

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -du -h /MarketingAnalytics 2>/dev/null
echo
127.0 K 127.1 K /MarketingAnalytics/RawData

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfs -df -h 2>/dev/null
echo
Filesystem      Size      Used Available Use%
hdfs://localhost:9000 228.3 G 148.0 K 21.5 G 0%

(base) patelkrina@patels-MacBook-Air-4 ~ % hdfs dfsadmin -report 2>/dev/null
echo
Configured Capacity: 245107195904 (228.27 GB)
Present Capacity: 23078664800 (21.49 GB)
DFS Remaining: 23078664800 (21.49 GB)
DFS Used: 151503 (147.95 KB)
DFS Used%: 0.00%
Replicated Blocks:
    Under replicated blocks: 1
    Blocks with corrupt replicas: 0
    Missing blocks: 0
    Missing blocks (with replication factor 1): 0
    Badly Distributed Blocks: 1
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0
Erasure Coded Block Groups:
    Low redundancy block groups: 0
    Block groups with corrupt internal blocks: 0
    Missing block groups: 0
    Badly Distributed Blocks: 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0

-----
Live datanodes (1):

Name: 127.0.0.1:9866 (localhost)
Hostname: localhost
Decommission Status : Normal
Configured Capacity: 245107195904 (228.27 GB)
DFS Used: 151503 (147.95 KB)
Non DFS Used: 222028439601 (206.78 GB)
DFS Remaining: 23078664800 (21.49 GB)
DFS Used%: 0.00%
DFS Remaining%: 9.42%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Fri Aug 07 14:27:49 IST 2026
Last Block Report: Fri Aug 07 13:20:22 IST 2026
Num of Blocks: 2

```

Tools /Technology used:

- Ubuntu (WSL)
- Hadoop 3.3.6
- OpenJDK 11+
- HDFS (Hadoop Distributed File System)
- Linux Terminal
- Nano Text Editor

Conclusion :

The practical successfully demonstrated the installation verification and configuration of Hadoop along with various HDFS operations. File management tasks such as creating directories, uploading, copying, moving, renaming, downloading, and deleting files were performed successfully. Hadoop monitoring commands were also used to verify cluster health, storage utilization, and running services, providing practical understanding of HDFS administration.